

CSS 430 A: Operating Systems

Spring 2021 | Tuesday/Thursday 8:45-10:45AM

Instructor: Yusuf Pisan (pisan@uw.edu)

- [Homepage](#) || [Research Group](#) || [All Course Evaluations](#) || [RateMyProfessors](#)

- **Office hours:** Mon 10-11am, Wed 3:30-4:30pm @ UW1-260Q - Signup via [Canvas Calendar](#)
- **Class Sessions:** 11:00-1:00 MW at UW1-020
- **Prerequisite:** CSS 343
- [Course website](#)
- You can submit [anonymous feedback](#) for this class anytime during the quarter

Course Description

This course introduces the logical design of operating systems. Topics covered include processes, threads, CPU scheduling, synchronization, deadlocks, memory management, virtual memory, file systems, I/O systems, protection, and security used in the popular desktop and server operating systems. A focus is made on the Unix operating system. The course utilizes Java in programming examples and as a basis for design.

Learning Objectives

ABET Outcomes:

This course contributes to the following ABET outcomes:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, , and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives

6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Textbooks

Operating System Concepts, <https://www.os-book.com/OS10/> 10th edition, Abraham Silberschatz, Peter Galvin, and Greg Gagne, Addison-Wesley, 2013. (available as an e-text)

Optional References

[*Understanding the Linux Kernel*](#), Daniel P. Bovet and Marco Cesati, O'Reilly, 2000

[*The Design of the Unix Operating System*](#), Maurice J. Bach, Prentice Hall, 1987

[*Advanced Programming in the Unix Environment*](#) 2nd Ed, W. Richard Stevens, Addison Wesley, 2005

Grading

- Exercises: 10%
- Homeworks: 20%
- Projects: 40%
- Midterm: 15%

- Final: 15%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. 2.0 roughly corresponds to 75%. Grades are not curved which means everybody can get a 4.0. Your success is not determined by how well or badly other students in the class are doing.

Inclusion

All students are welcome in CSS 430 and are entitled to be treated respectfully by both classmates and the course staff. I strive to create a challenging but inclusive environment that is conducive to learning for all students. If at any time you feel that you are not experiencing an inclusive environment, or you are made to feel uncomfortable, disrespected, or excluded, please report the incident so that we may address the issue and maintain a supportive and inclusive learning environment. You may contact me or the [CSS Advisors](#) to express your concerns. Should you feel uncomfortable bringing up an issue with a staff member directly, you may also consider sending [anonymous feedback](#).

Policies

Assignment Submission: You can miss one exercise without penalty.

For some homeworks and projects, you will be able to work individually or in pairs. See homework/project description on whether pair-work is allowed or not. When working as a pair

- Both students must enter a comment in the Canvas assignment with their name and their partner's name 72 hours before the assignment is due. Submit an empty file if necessary to leave a comment.
- All substantial work must be performed with both students present (in-person or virtually).

You can submit one and only one project 24 hours late without any penalty (if working as a pair both people will need to use their one-time extension). If you are using your 24 hour extension, you must post a brief explanation to "Using Extension" assignment before the assignment due date.

If you have any questions about your assignment grade, you must contact me within one week.

Plagiarism: Other than when working in pairs, talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct:

<http://www.washington.edu/cssc/for-students/student-code-of-conduct/> I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Any code or text that you copy from an external source must have a comment (or reference) indicating its origins.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Getting Help: There are multiple ways of getting help:

- **Discord**

With an active class, discord can be one of the easiest and fastest ways to get help. Discord channel is an extension of our class and the same rules for in-class behavior apply to discord. I will also monitor the discord channel and contribute to the discussion from time to time. When asking questions:

- **Do not** post code.
- Be clear in stating the problem. Explain how you tried to solve it. Make sure the answer is not the first link that comes up in a Google search.
- You can post a warning/error message from compiler and ask how others have resolved it.
- You can ask how a specific C++ function gets used
- You can discuss your approach in high-level terms
- If you post a question and then figure out the answer, post your answer so it benefits others in the class

In general, if you can discuss the problem in English without having to show code, you can discuss it on Discord.

- **Office Hours**

Make an appointment to come and see me during office hours. I will not debug your code, but I will answer questions and try to help you figure out the problem. I will always ask what you have already done so far to solve the problem, so come prepared.

If you have a personal matter, please visit me during office hours to talk.

- **Email**

Email is not an effective way to get help. Unless your question is of personal nature, it should be posted on discord. If you believe the question really needs my attention, you can tag me on discord.

Extenuating Circumstances

I recognize that our students come from varied backgrounds and can have widely-varying circumstances. If you have any unforeseen circumstances that arise during the course, please do not hesitate to contact me to discuss your situation. **The sooner I am made aware, the more easily I can provide accommodations.** While some amount of “productive struggle” is healthy for learning, you should ask me for help if you have been stuck on an issue for a very long time.

Life happens! While our focus is providing an excellent educational environment, our course does not exist in a vacuum. My ultimate goal is to provide you with the ability to be successful, and I encourage you to work with me to make that happen. Please don't suffer in silence.

Access and Accommodations

Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course.

If you have not yet established services through DRS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are welcome to contact DRS at 425.352.5307 or rosal@uw.edu.

DRS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions. Reasonable accommodations are established through an interactive process between you, your instructor(s) and DRS. It is the policy and practice of the University of Washington to create inclusive and accessible learning environments consistent with federal and state law.

For Our Veterans

Welcome! We at UW Bothell understand that the transition into civilian life can be challenging for our veteran students and we have many resources for any who may want to reach out for guidance or assistance. This includes our Vet Corp Navigator through the WDVA and our Student Veterans Association (SVA). Please contact Veteran Services at 425.352.5307 or rosal@uw.edu. For those of you needing more URGENT support, please call The Suicide Prevention Hotline 1.800.273.8258 or connect with the UWB CARE Team <https://www.uwb.edu/studentaffairs/care-team>.

Religious Accommodations

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at [Religious Accommodations Policy](#). Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form](#).

Common Course Policies for the School of STEM

See https://bit.ly/STEMClassroom_Policies for additional information on Academic Integrity, Access and Accommodations, Classroom Emergency Preparedness, For Our Veterans, Grade of Incomplete, Inclement Weather, Parenting Resources, Respect for Diversity, Student Support Services, Wondering How to Address Faculty? etc. (P.S. I prefer to be addressed as 'Professor Pisan')